

Applied Human Computer Interaction (AHCI)

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Chapter 10 – Universal Design

Universal Design

Universal Design – What & Why: Designing for *real people*, not a “typical user”

Users differ in:

- Abilities (visual, motor, cognitive)
- Culture, language, age, experience

Goal: Include as many users as possible

Key idea: Don’t design just for *yourself*

Example: Not everyone uses apps the same way—some prefer voice notes, others text, some rely on captions.

The Problem with Average Users

- No such thing as a “standard user”
- People vary in:
 - Tech skills
 - Attention span
 - Devices (low-end phones vs flagship)
- Risk: Excluding users unintentionally

Example: A heavy app may work fine on iPhones but lag badly on budget Android phones → exclusion.

What is Universal Design?

- Designing systems usable by:
 - Everyone
 - Across different situations

Two approaches:

- **Redundancy** (multiple ways to interact)
- **Assistive compatibility** (screen readers, etc.)

Example: Videos with **captions + audio** → usable in noisy places or for hearing-impaired users

Principles of Universal Design

- Equitable Use
- Flexibility in Use
- Simple & Intuitive
- Perceptible Information
- Tolerance for Error
- Low Physical Effort
- Size & Space for Use

Think of this as a **designer's checklist**

Principles of Universal Design

Principle 1 – Equitable Use

- Same or equivalent experience for all users
- Avoid exclusion or stigma

Example: Login via:

- Email
- Google
- Fingerprint → Multiple options for different users

Principle 2 – Flexibility in Use

- Supports different preferences & abilities
- Allows customization

Example: Playback speed control (1x, 1.5x, 2x) on videos → Gen Z often prefers faster consumption

Principles of Universal Design

Principle 3 – Simple & Intuitive

- Easy to understand regardless of experience
- Minimal complexity
- Clear structure & feedback

Example: Short-form apps like TikTok-style navigation: swipe up = next video → No instructions needed

Principle 4 – Perceptible Information

- Information presented in multiple formats
- Works across environments & abilities

Example: Notifications:

- Sound
- Vibration
- Visual pop-up → Ensures user doesn't miss it

Principles of Universal Design

Principle 5 – Tolerance for Error

- Prevent mistakes or reduce damage
- Allow recovery

Example: “Undo send” in email or delete confirmation prompts
→ Saves users from embarrassing mistakes

Principle 6 – Low Physical Effort

- Easy, comfortable interaction
- Reduces fatigue

Example: Auto-fill forms & one-tap payments → No repetitive typing

Principle 7 – Size & Space for Use

- Accessible regardless of body size or mobility
- Easy to reach & interact

Example: Large buttons & thumb-friendly design (important for mobile users)

Usable Senses

The 5 senses (sight, sound, touch, taste and smell) are used by us every day

- each is important on its own
- together, they provide a fuller interaction with the natural world

Computers rarely offer such a rich interaction

Can we use all the available senses?

- ideally, yes
- practically – no

We can use • sight • sound • touch (sometimes)

We cannot (yet) use • taste • smell

Multi-Modal Interaction

- Using **multiple senses** to interact with systems
- Key idea: Don't rely on just one channel (like visuals)
- Supports **universal design through redundancy**

Example: Watching videos with **audio + captions + visuals** at the same time.

Multi-Sensory Interaction:

- Real-world interaction uses multiple senses
- Systems should mimic this
- Leads to: Richer experience and more natural interaction

Example: Gaming → visuals + sound effects + controller vibration

Multi-Modal vs Multi-Media

Often confused terms in HCI

Key difference:

- **Multi-modal** → senses (how we perceive)
- **Multi-media** → content formats (how info is presented)

Simple Idea: Modal = *human senses* - Media = *type of content*

What is Multi-Modal Interaction?

- Uses **more than one sensory channel**
- Focus on *how users interact*

Examples of modes: Visual, Auditory, Touch

Example: A text editor that: Displays text (visual) and Reads it aloud (audio)

Multi-Modal vs Multi-Media

What is Multi-Media? Uses **different types of media/content** .
Focus on *how information is presented*

Types of media: Text, Images, Video, Animation, Audio

Multi-Media – Example E-learning system: Videos, Animations, Text, Images All mostly **visual**, so not necessarily multi-modal

Multi-Media: Instagram posts: Images + reels + text captions

YouTube videos: Video + thumbnails + overlays

Different media, but mainly visual + some audio

Speech as an Interaction Mode

- Natural and powerful communication method
- Humans are highly skilled at speaking & understanding
- Seems simple → actually very complex

Example: Using voice notes instead of typing long messages

Structure of Speech Speech is built in layers:

Phonemes → Allophones → Morphemes

Like building blocks of language

Phonemes (Basic Units)

- Smallest units of sound
- Around **40 in English**
- No meaning on their own

Example: Sounds like “b”, “p”, “t”

Allophones (Sound Variations)

- Variations of phonemes depending on context
- Total sounds: **120–130**
- Same phoneme, slightly different pronunciation

Example: “t” in: “top” vs “stop” → sounds slightly different

Speech

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Other Key Speech Concepts

Prosody → tone, pitch, stress, pauses

Co-articulation → sounds change based on context

Syntax → sentence structure

Semantics → meaning

Prosody – Emotion & Meaning

- Adds feeling and emphasis
- Changes meaning without changing words

Example: “I didn’t say that” → different meanings based on stress

Other Key Speech Concepts

- Sounds influenced by surrounding sounds
- Makes speech fluid but harder for machines

Example: Fast speech → words blend together

Syntax vs Semantics

- **Syntax** = grammar structure
- **Semantics** = actual meaning

Example: “Can you pass the salt?”

→ Syntax: question

→ Semantics: request

Speech Recognition Challenges

People speak differently: Accent, Tone, Speed, Volume

Example: Slang + mixed languages (e.g., Urdu + English in one sentence)

More Challenges Same meaning, different sentences:

- “Turn off the light”
- “Switch the light off”
- Background noise interference
- Fillers: “umm”, “uhh”, “like”

Understanding Meaning is Hard

- Words alone are not enough
- Systems need: Context, Intent, Background knowledge

Example: “I don’t like this” → What is “this”?

The Phonetic Typewriter

- Developed for **Finnish** (phonetic language: written as spoken)
- Uses **neural networks** to recognize speech
- Trained on **one speaker**, generalizes to others

Example: Imagine talking to a speech-to-text app on your phone that types what you say in real-time

How It Works

- Neural network clusters **similar sounds**
- Each cluster → labeled with corresponding **character**
- When speaking: System finds closest sound cluster - Prints the character
- Requires **large dictionary** for minor variations

Analogy: Like predictive typing in texting, but for sounds → letters

Speech Synthesis

Definition: Generation of artificial speech

- Provides information in a **natural and familiar way**
- Complements other interaction modes

Example: Voice assistants like Siri, Alexa, or Google reading notifications aloud

Challenges of Speech Synthesis

- Similar to speech recognition: **Prosody** (intonation, stress, pauses) is hard to generate naturally
- **Additional issues:**
 - Intrusive → may require headphones or cause noise
 - Transient → information is **harder to review** than text

Example: Voice reading a long email → easy to miss details compared to skimming text

When Speech Synthesis Works Best

- Effective in **constrained applications** where:
 - User is motivated to overcome limitations
 - Few alternatives exist

Key applications:

- **Screen readers** → for visually impaired users
- **Warning signals** → pilots, drivers, or critical alerts

Example:

Voice alerts in apps like Google Maps or Apple CarPlay
→ frees visual attention while multitasking

Non-Speech Sounds

- Sounds that are **not speech**: boings, bangs, squeaks, clicks
- Commonly used for: Warnings, Alarms , Feedback in interactions

Example: Game sound effects → jumps, hits, level-ups Keyboard clicks → improve typing accuracy

Benefits of Non-Speech Sounds

- **Language & culture independent** → works globally
- Improves **accuracy and engagement**
- Supports **dual-mode displays**:
 - Information in **two sensory channels** → visual + auditory
 - Resolves ambiguity

Example: Mac OS feedback sounds: File selection, Error , Disk insert → Users quickly recognize actions without reading

Best Use Cases

- **Transient information** → quick alerts
- **Background status** → ongoing system feedback
- Redundant channel for **unambiguous communication**

Example: Notification sounds in apps like WhatsApp, Instagram → conveys info without looking at phone

Auditory Icons

- Use **natural sounds** to represent objects or actions. Map real-world **semantics** onto digital actions

Example:

- Trash action → smashing glass
- Folder → papery rustle
- Copying a file → liquid pouring sound
- Additional cues: Muffled sounds → object obscured or in background. Stereo positioning → direction of sound

Benefits of Auditory Icons

- Intuitive → users **recognize actions quickly**
- Adds **redundancy** → visual + auditory cues
- Can convey **progress** or **size**: Rising pitch → progress of copy. Louder sound → bigger file

Example: Video games often use sound cues for action feedback

- Chest opening → creak
- Sword swing → metallic clang

Earcons

Synthetic sounds used to convey information

- Built from **structured combinations of notes (motives)**
- Can represent: Actions & Objects

Example: App notifications with unique tones (like different alert sounds in Slack or Discord)

Types of Earcons

- **Compound earcons:** Multiple motives combined → more complex sound
- **Family earcons:**
 - Similar earcons → similar actions or objects
 - Example: “error” sounds across different OS notifications

Example: Different error tones for Wi-Fi disconnect, low battery, or message failed → same family, recognizable

Touch in HCI – Haptic Interaction

- Interaction through the **sense of touch**
- Two main types:
 - **Cutaneous perception** → skin sensations (vibration, pressure)
 - **Kinesthetics** → movement, position, force feedback

Example: Phone vibration when typing or receiving notifications

What Touch Can Communicate

- Shape & texture
- Resistance (force feedback)
- Temperature
- Spatial awareness

Example: Gaming controllers → vibration when hit or during explosions

Haptic Technologies: Electronic Braille displays → tactile reading for visually impaired

Handwriting Recognition

- Converts **handwritten input into digital text**
- Natural, everyday communication method
- Captured using: Stylus + touchscreen, Digitizing tablets

Example: Writing notes on tablets like iPads or stylus-based Android devices

Challenges in Handwriting Recognition

- Personal writing styles vary
- Letters connect (co-articulation)
- Messy or fast writing = harder to interpret

Example: Your handwriting vs your friend's → completely different shapes

Improvements & Current State

- Systems now: Use **stroke-based recognition** (not just images). Special simplified alphabets (e.g., Graffiti style input). Most people still type faster than they write on screens

Gesture in HCI

- Interaction through **physical movement**
- Includes:
 - Pointing
 - Hand movements
 - Body gestures

Example: Swiping on phones (left/right/up) to navigate apps

Applications of Gesture

- **Gestural input** → e.g., “put that there” systems
- **Sign language** → communication for hearing-impaired users

Example: Video calls where users use hand gestures to communicate.

Gesture Technology

- **Data gloves** → capture hand movement
- **Position sensing systems** (e.g., smart rooms, motion sensors)

Example: AR/VR systems detecting hand gestures without controllers

Gesture in HCI

- Natural and intuitive interaction
- Faster than typing in some contexts
- Enhances communication across users

Example: Pointing to select items on touchscreen instead of typing

Challenges of Gesture

- User-dependent (everyone gestures differently)
- Variability & ambiguity
- Co-articulation (gestures blend together)

Example: Different meanings of the same gesture across cultures

Designing for Users with Disabilities

Goal: Inclusive interaction for all users. Use multi-modal approaches: Audio, Visual, Touch, Gesture.

Visual Impairment. Tools: Screen readers. Audio-based systems

Example: Phones reading out screen content for navigation

Hearing Impairment. Solutions: Text communication, Captions, Gesture/sign language.

Example: Watching videos with subtitles (very common behavior)

Physical Impairment: Alternatives to traditional input:

Speech input/output, Eye-gaze tracking, Gesture systems, Predictive keyboards

Example: Auto-complete typing to reduce effort

Designing for Users with Disabilities

Speech Impairment: Solutions: Speech synthesis & Text-based communication

Example: Typing messages that are read aloud by devices

Cognitive Differences (Dyslexia & Autism)

Dyslexia: Speech input/output helps reading & writing

Autism: Tools for communication & structured learning

Example: Apps that read text aloud or simplify content

Designing for Users with Disabilities

Age Groups & Design

- **Older adults:**

- Accessibility tools
- Memory aids
- Social communication tools

- **Children:**

- Simple interfaces
- Engaging input/output
- Should be involved in design

Example: Large buttons for seniors vs interactive learning apps for kids

Designing for Users with Disabilities

Cultural Differences in Design

- Users differ by: Nationality, Language, Gender, culture, beliefs
- Affects: Symbols, Colors, Gestures

Examples: gesture → positive in some cultures, offensive in others.

Colors → red = danger in some places, celebration in others